Importance of N_2H_x Reaction Pathways to Optimize Okafor Reduced Mechanism for NH_3/H_2 Flame Chemistry

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Research Background

Traditional sources of energy rely on combustion of fossil fuels that produce carbon emissions which have adverse effects on the environment:

- Global Warming
- Climate Change

New carbon-free alternative fuels are needed to prevent exacerbating the effects of carbon pollutants

- The Japanese government seeks to reduce total greenhouse gas emissions by 46% before 2030
- By 2050, Japan envisions its cities to be carbon-neutral

Research Background

Ammonia, NH_3, is gaining interest among researchers for its potential to meet energy needs in a variety of applications.

Ammonia is a suitable fuel for combustion application because:

- Ammonia combustion produces no CO_2
- Ammonia has a high volumetric energy density
- Easier to store than hydrogen
- Infrastructure to produce and transport ammonia already exists
- Used as H_2 carrier

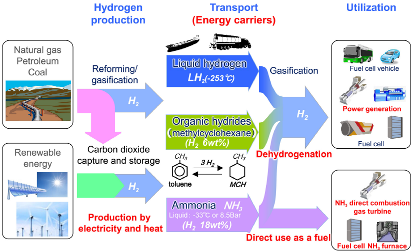

Figure: Hideaki Kobayashi et al. (Jan. 1, 2019). "Science and Technology of Ammonia Combustion". In: *Proceedings of the Combustion Institute* 37.1, pp. 109–133. ISSN: 1540-7489. DOI: 10.1016/j.proci.2018.09.029

Objectives

- Validate Abe Mechanism, at varying combustion conditions, for fuels:
  - NH_3/H_2/Air
  - CH_4/NH_3/Air
- Flame properties being evaluated:
  - Laminar burning velocity
  - Specie profiles
  - Intermediate species profiles
- Optimize reaction mechanism to fit experimental data for all conditions above satisfactorily

Prior Findings

- Validate Abe Mechanism, at varying combustion conditions, for fuels:
  - NH_3/H_2/Air
  - CH_4/NH_3/Air
- Flame properties evaluated:
  - Laminar burning velocity

Prior Findings — NH3/H2/Air

Abe mechanism reproduces LBV of NH/H2/Air flame to match experimental data

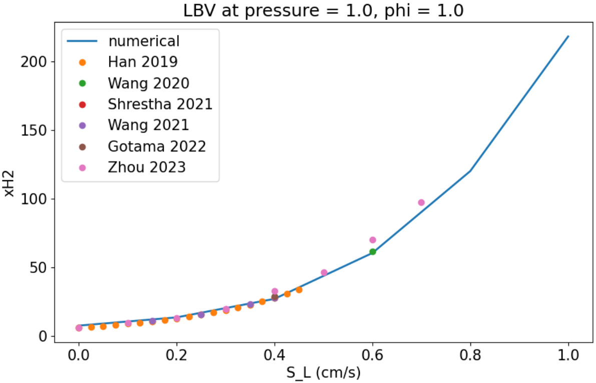

Figure: Laminar burning velocities of NH3/H2/Air flames at varying H2 concentrations while pressure and equivalence ratio is kept constant at 1atm and 1.0 respectively

Plotting of experimental data from Okafor 2019[1] was corrected

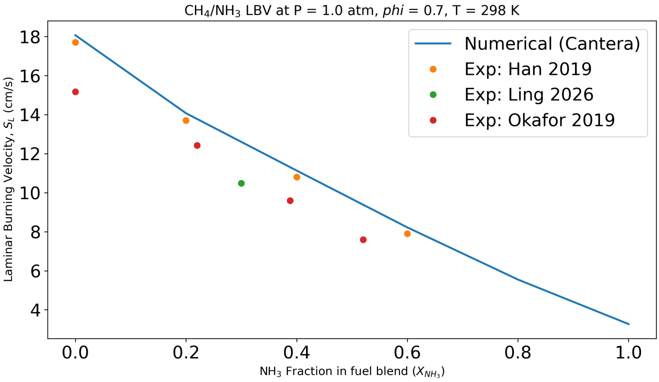

Figure: Laminar burning velocities of CH4/NH3/Air flames at varying NH3 concentrations while pressure and equivalence ratio is kept constant at 1atm and 0.7 respectively

[1] Ekenechukwu Chijioke Okafor et al. (June 1, 2019). "Measurement and Modelling of the Laminar Burning Velocity of Methane-Ammonia-Air Flames at High Pressures Using a Reduced Reaction Mechanism". In: *Combustion and Flame* 204, pp. 162–175. ISSN: 0010-2180. DOI: 10.1016/j.combustflame.2019.03.008.

Prior Findings — Failed Conditions

- LBV overprediction by Abe mech observed across most CH4/NH3/Air conditions.
  - Pressure $= 1$ atm, $\phi = 1.4$
  - Pressure $= 1$ atm, $\phi = 1.6$
- Experimental data for higher pressures, 3atm and 5atm, are lacking for both fuels
- Numerical simulations with Okafor reduced mechanism failed to converge
  - Unable to compare LBV from Okafor Reduced Mechanism and Abe mech at the conditions discussed

Next Steps

- Validating Abe mech's specie profiles with experimental data from literature
- Speciation data for relevant fuels and corresponding fuels selected
  - CH_4/NH_3: NH_3, NO, CO
  - NH_3: OH, NH, NO
  - NH_3/H_2: NH_2, NH, NO_x
- For each speciation dataset, simulation methodology employed in the corresponding paper must be used for validating Abe mech

Specie Profiles — CH4/NH3

- From Mendiara and Glarborg 2009, the speciation data is used to validate NH3, NO and CO profiles
- Plug Flow Reactor
  - Diameter = 12 mm
  - Length = 120 cm
  - Gas flow rate = 1 dm^3/min
  - Temperature = 973-1773 K

Table: Experimental conditions that need to be simulated with the plug flow reactor. Teresa Mendiara and Peter Glarborg (Oct. 1, 2009). "Ammonia Chemistry in Oxy-Fuel Combustion of Methane". In: *Combustion and Flame* 156.10, pp. 1937–1949. ISSN: 0010-2180. DOI: 10.1016/j.combustflame.2009.07.006

Run	CH_4/ppm	O_2/ppm	NH_3/ppm	N_2/%	ϕ	Residence time/s
1	2515	3479	484	99.35	1.55	1296/T (K)
2	2513	5036	468	99.20	1.07	1296/T (K)
3	2508	40062	468	95.69	0.13	1293/T (K)

Specie Profiles — NH3

- From Brackmann et al. 2016, speciation data of OH, NH and NO profiles are used to validate Abe mech.
- Stagnation Flame Reactor
  - Steel disk to stabilize the flame
  - Flame stabilized at $\sim$6 mm above burner surface so domain should be $>$ 6 mm
  - Temperature profiles from the experiment used as input
  - GRAD parameter was set to 0.02

Table: Experimental conditions that need to be simulated with the stagnation flame reactor. Christian Brackmann et al. (Jan. 1, 2016). "Structure of Premixed Ammonia + Air Flames at Atmospheric Pressure: Laser Diagnostics and Kinetic Modeling". In: *Combustion and Flame* 163, pp. 370–381. ISSN: 0010-2180. DOI: 10.1016/j.combustflame.2015.10.012

Flame	NH_3 (L_n/min)	Air (L_n/min)	N_2-flame (L_n/min)	N_2-coflow (L_n/min)
$\phi = 0.9$	2.62	10.3	0	1.2
$\phi = 1.0$	3.38	12.1	0.87	1.2
$\phi = 1.2$	3.9	11.6	0	1.2

Specie Profiles — NH3/H2

- From Alturaifi, Mathieu, and Petersen 2023, speciation data of NH3, NO, NOx, N2O and NH2 profiles used.
- Premixed Laminar Flame-Speed Reactor/Burner-stabilized flat flame
  - Domain/width of $\sim$ 20 mm
  - Experimental flame's temperature profile used as input

Table: Experimental conditions that need to be simulated. Sulaiman A. Alturaifi, Olivier Mathieu, and Eric L. Petersen (Jan. 1, 2023). "A Shock-Tube Study of NH3 and NH3/H2 Oxidation Using Laser Absorption of NH3 and H2O". In: *Proceedings of the Combustion Institute* 39.1, pp. 233–241. ISSN: 1540-7489

Flame	X_{NH_3}	X_{H_2}	X_{O_2}	X_{Ar}	V_0 (cm/s)	Φ	p (mbar)
I	0.25	0.05	0.21	0.48	35.1	1	60
II	0.24	0.07	0.21	0.47	42.1	1	50
III	0.22	0.10	0.21	0.46	42.1	1	50
IV	0.21	0.13	0.21	0.45	42.0	1	50
V	0.22	0.10	0.24	0.43	42.2	0.9	50
VI	0.22	0.10	0.20	0.48	42.5	1.1	50
VII	0.25	0.05	0.21	0.48	23.4	1	90
VIII	0.25	0.05	0.21	0.48	17.6	1	120

Thank You